Jahangirnagar University
জাহাঙ্গীরনগর বিশ্ববিদ্যালয়

Department of Computer Science and Engineering

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Course Title: Digital Logic Design Laboratory

Course Code: CSE-254

Experiment No: 04

Experiment Title: Design and implementation of a full Adder using 2 Half adders and Converting binary-gray-binary code.

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Experiment No: 04

Experiment Name:

Implementation of the full Adder using two half adders and converting binary-gray code

Objectives:

- To design and verify half and full adder circuits using logic gates.
- To convert binary to gray code.

Required Components

Digital trainer board, bread board, connecting wires, IC 7408, IC 7432, IC 7486.

Theory

Half Adder adds two 1-bit binary numbers. and full adder adds three 1-bit binary numbers.

To convert gray to binary X-OR Gate is used to the input bits and vice-versa output

bits are X-OR operated.

Circuit Diagram

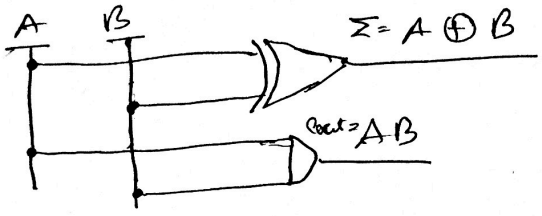


Fig: Half Adder

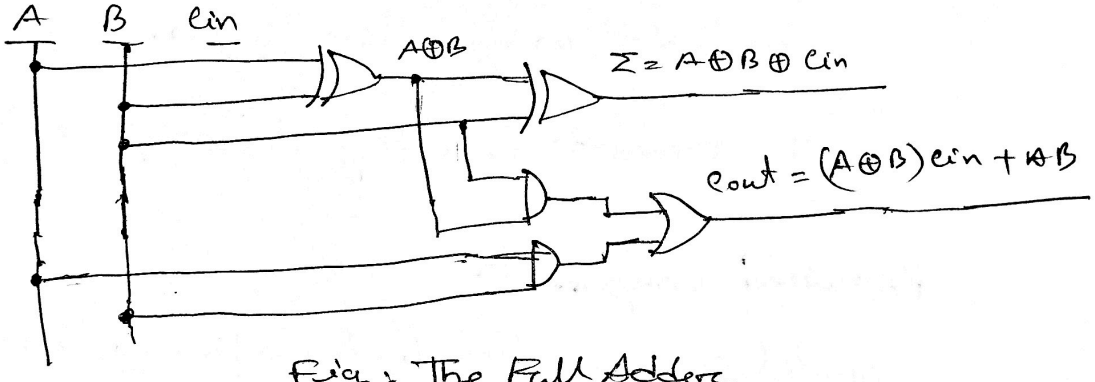


Fig: The Full Adder

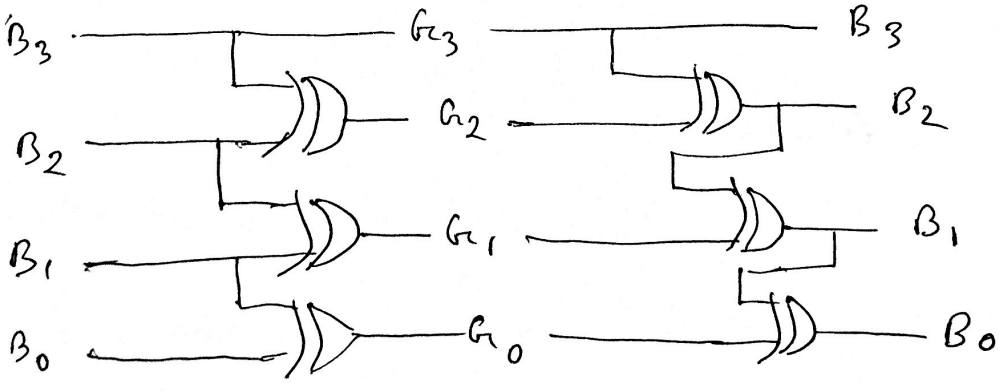


Fig: Binary - Gray - Binary conversion.

Truth Tables:

The half Adder Table:

A	B	Σ	Cout
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

The full Adder Table:

A	B	Cin	Σ	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

The 'Binary - Gray - Binary Conversion Table:

Binary	Gray	Binary
0000	0000	0000
0001	0001	0001
0010	0011	0010
0011	0010	0011
0100	0110	0100
0101	0111	0101
0110	0101	0110
0111	0100	0111
1000	1100	1000
1001	1101	1001
1010	1111	1010
1011	1110	1011
1100	1010	1100
1101	1011	1101
1110	1001	1110
1111	1000	1111

Result:

- For half and full adder output matched the truth tables.
- Sum, carry LED lit correctly
- Binary to Gray conversion verified all combinations.

Conclusion:

- Successfully implemented Half Adder, full Adder, and code conversion using gates
- Verified functionality via truth tables.
- Demonstrated the importance of XOR gates.